Forecasting small-business AI use: data, baseline, and reliability

ISYE 4031 — Milestone 2 — Group 1 — October 30, 2026
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1. Question and response to feedback

Question. Can a simple regression forecast an industry-by-size group's AI-use rate three Census releases (about six weeks) ahead more accurately than naive benchmarks, and is it reliable enough to guide near-term outreach by small-business support organizations? **Target:** the published percentage of businesses in a group reporting AI use in the last two weeks (BTOS Q7). Small business means employers with 1–99 employees.

Changes. TA feedback during Milestone 1 asked us to fix the target and model, add predictors, and run a baseline. Milestone 1 showed that neither expected use (Q24) nor business performance reduced error, so we dropped performance and kept expected use as one tested addition. The focus is now reliability: whether any gain holds across dates and size bands, and how large errors are relative to the survey's own noise. **Hypothesis tested:** the regression has lower mean absolute error (MAE) than the six-release average at most forecast dates and in every size band.

2. Data and preparation

Source. Public Census Business Trends and Outlook Survey (BTOS) sector-by-size workbook, accessed October 3, 2026 and frozen with a SHA-256 hash [1]; a copy downloaded October 20 (renamed by Census) supplies one newer release for a prospective check. One observation is a NAICS sector × size band in one biweekly release, not a firm. Variables: AI-use rate and expected-use rate (%), the rate's standard error (percentage points, pp), sector, size band (1–4, 5–9, 10–19, 20–49, 50–99) and release index. We use the 22 releases from the November 17, 2025 question revision through the September 24, 2026 publication; earlier rates measured a narrower question and are not comparable [2].

Pipeline step (all in `analysis.py`)	**Count**
Published cells: 20 sectors × 5 size bands × 22 releases	2,200
Suppressed by Census (S), kept as missing	498 (22.6%)
Groups kept: exclude multi-sector XX; require current and expected use in ≥7 of the first 8 releases	71 groups
Issue–target pairs (release t → t+3); complete cases	1,349; 1,335
Scored forecasts: 9 issue dates, April 23 – August 13, 2026	635

Quality. No duplicate group-release keys; all rates lie in 0–100. Suppression is similar across bands (20–25%, Table 1), but we do not impute: suppressed cells are small or unreliable estimates, so imputing them would invent precision. Screening drops sparse sectors, so results describe 71 well-measured groups, not all small businesses. Survey noise rises sharply with firm size because fewer firms are sampled in larger bands: the median standard error is 1.7 pp for 1–4 employees and 5.5 pp for 50–99.

Dependence and leakage. The same groups recur every release, BTOS rotates panels, and it replaced its sample in July 2026 [3], so the 635 forecasts share nine origins and common shocks and are not independent. A forecast is issued on the publication date of release t and uses only values published by then. A training pair enters only after its target is published, and screening uses only releases before the first issue date; the code asserts both. Outliers were investigated, not deleted (Section 4).

3. Exploratory findings

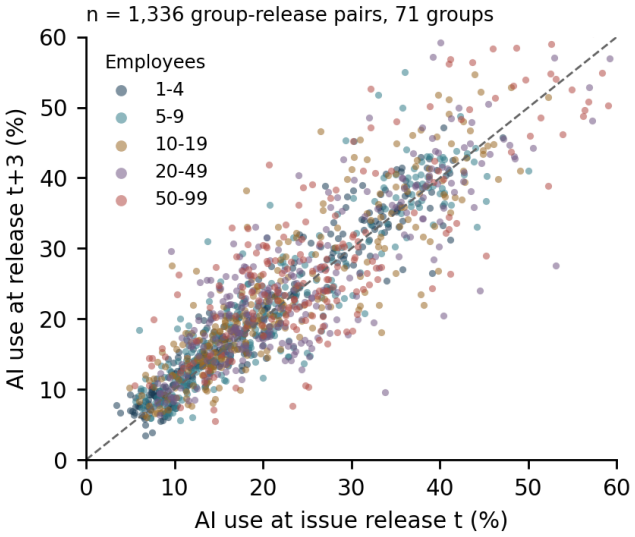


Figure 1. Each group's AI-use rate at issue release t against its rate three releases later. Dashed line: no change.

Employees	Cells	Suppressed %	Median rate %	Median SE (pp)
1–4	440	22.5	16.3	1.7
5–9	440	23.9	18.4	2.4
10–19	440	22.0	20.1	2.8
20–49	440	19.8	22.5	3.6
50–99	440	25.0	26.4	5.5
All	2,200	22.6	20.9	3.0

Table 1. Data quality by size band, all 20 sectors and 22 releases. SE: published standard error of the AI-use rate.

Figure 1 shows that a group's current rate is the main signal: points follow the no-change line. Scatter around the line widens at higher rates and is widest for 50–99 employees, so a single pooled error is misleading and we report errors by size band. **Table 1** shows that much of that scatter is sampling noise: a 50–99 estimate can move by about 11 pp (two standard errors) with no real change. A model cannot forecast below this noise floor, so we judge its errors against the target's standard error.

4. Baseline and initial results

Model. Pooled ordinary least squares for group g at issue release t:

$$y_{g,t+3} = \beta_0 + \beta_1 y_{g,t} + \text{industry effects} + \text{size effects} + \beta_2 t + \varepsilon_{g,t+3}$$

The target is a continuous percentage. Industry and size effects capture stable level differences, $y_{g,t}$ captures persistence, and t captures the common upward trend; every term is known on the issue date. **References:** carry the latest rate forward, and the mean of the group's latest six releases. **Validation:** expanding window over nine issue dates, horizon three releases; the model is refit at each date on pairs whose targets were already published (1,124 rows at the last date). **Metric:** MAE in pp; root mean squared error (RMSE) penalizes large misses.

Forecast	MAE	RMSE	1–4	5–9	10–19	20–49	50–99
Carry latest rate forward	4.18	5.79	2.32	3.67	3.76	5.20	6.65
Mean of latest six releases	3.40	4.65	2.48	3.19	3.38	3.63	4.66
Regression (baseline)	**3.33**	**4.60**	**2.06**	**3.37**	**2.78**	**3.54**	**5.47**
Regression + expected use	3.33	4.61	2.07	3.37	2.79	3.57	5.45

Table 2. Out-of-sample error (pp), 635 forecasts at 9 issue dates; size columns are MAE by employee band (97–135 forecasts each).

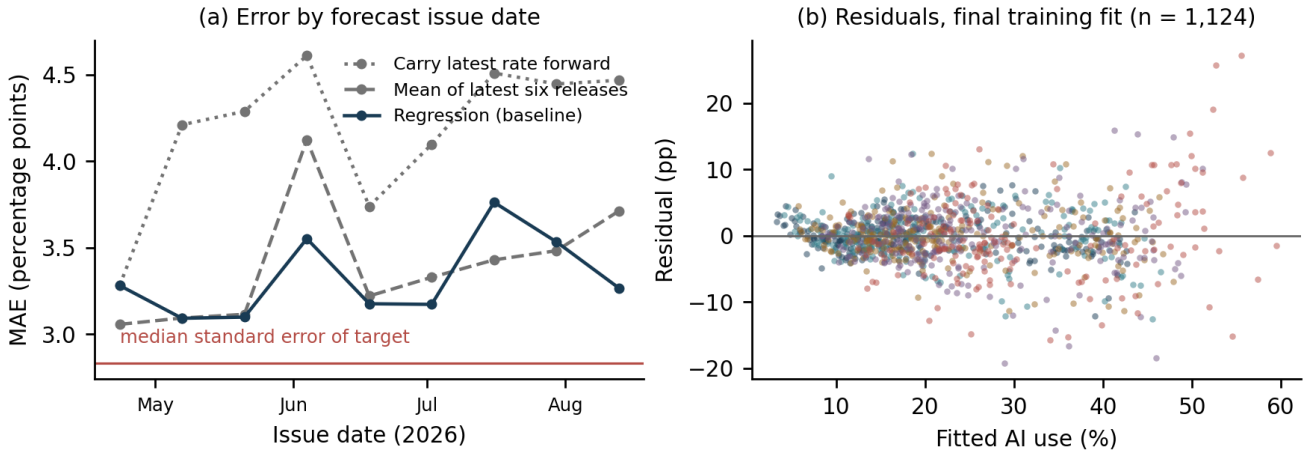


Figure 2. (a) MAE at each issue date; red line, median standard error of the targets (2.83 pp). (b) Final-fit residuals against fitted values, colored by size band as in Figure 1.

Results. The regression beats carrying the rate forward by 0.85 pp but beats the six-release mean by only 0.07 pp, and at only 6 of 9 dates (Figure 2a). It is better for 1–4 and 10–19 employees and worse for 5–9 and 50–99 (5.47 vs. 4.66 pp), so the hypothesis is **not supported**. The six-release mean under-predicts by 2.05 pp on average because adoption is rising; the regression's trend term removes that bias (+0.11 pp). Expected use changes nothing. The current-rate coefficient is 0.21 at the last fit, so once industry and size are known, the latest reading adds little; the model behaves like a smoothed group average plus a trend. This matches Table 1: much of each release-to-release change is sampling noise.

Diagnostics. Residuals fan out as fitted rates rise (Figure 2b), so errors are not constant across groups. The largest residuals (up to 27 pp) and the four highest Cook's distances all belong to Information firms with 50–99 employees. No point is highly influential (maximum Cook's distance 0.08). Refitting without the five most influential rows raised last-date MAE from 3.27 to 3.37 pp, so we keep them. 84% of regression errors are smaller than two standard errors of their target, and absolute error correlates with the target's standard error (r = 0.44).

Prospective check. Release 202620, published October 8, appeared after the Milestone 1 specification was frozen. We forecast it exactly as above, from the August 27 issue date. Census revised none of the earlier values, and screening selects the same 71 groups.

Forecast (71 groups, one issue date)	MAE	Bias	MAE, 50–99
Carry latest rate forward	3.90	−1.74	6.75
Mean of latest six releases	3.34	−2.20	4.58
Regression (baseline)	**3.09**	**−0.30**	**3.95**
Regression + expected use	3.10	−0.39	3.95

Table 3. Error (pp) on the first release published after the model was frozen; median target standard error 2.68 pp.

The regression leads by 0.25 pp here and also wins for 50–99 employees, and the six-release mean again under-predicts. One issue date is too few to overturn the historical result, but it is the first evidence not reused from model development.

5. Preliminary answer, limitations, and next steps

Preliminary answer. Not yet. A simple regression is a sound baseline: it clearly beats no-change forecasts and is near the survey's noise floor for the smallest firms. However, its historical edge over a six-release average is small and inconsistent across dates and size bands, and it fails for 50–99 employees. The first unseen release favors the regression (3.09 vs. 3.34 pp), but one release cannot settle the question. For outreach planning, a recent average performs about as well and is easier to explain. In this setting, deployment is limited more by measurement noise than by model choice.

Limitations. (i) Failure mode: forecasts for the largest band, especially Information, carry errors above 5 pp, comparable to real differences between groups. (ii) Mostly reused data: the nine historical dates were already inspected in Milestone 1 and are dependent; only one release so far is truly unseen. (iii) Association, not causation: industry and size effects describe where AI use is higher, not why, and say nothing about the effect of outreach. (iv) Performance is not readiness: 22 releases cannot show seasonal stability, and the July 2026 sample replacement may shift levels.

Most important next step (Milestone 3). Keep scoring the frozen models prospectively as releases publish (202621 on October 22, 202622 on November 5) with `prospective.py`. Compare machine-learning models (regularized regression with lags, gradient boosting) under the same expanding-window protocol. Judge gains against the standard error, per size band, with a paired test across issue dates.

6. Team contributions and reflection

Contributions. Rahul Mehta, Ishaan Mohan, Zhengyu Chris Liu, Timothy Kim and Chau M Duong contributed approximately equally to the Milestone 2 work. Rahul assembled the report and analysis package.

What we learned and what changed. The question shifted from which predictors help to whether any gain exceeds the survey's own noise. Expected use and business performance did not help, so we dropped performance and kept one tested addition. Reporting errors by size band revealed a failure that pooled MAE hid. Outliers looked like errors at first, but removing them made forecasts worse.

Feedback and response. Milestone 1 meetings: the TA asked for a fixed target and model, predictors, a baseline, and attention to measurement and short history. We responded with the equation, references, and standard-error comparison above. Graded Milestone 1 feedback: pending. Meeting 1 (October 20) feedback and response: pending. Meeting 2 (October 27) feedback and response: pending.

Next-stage owners. Prospective scoring on new releases: pending. Machine-learning models: pending. Per-band, noise-adjusted evaluation and paired test: pending. Final report integration: pending.

References. [1] U.S. Census Bureau, BTOS data downloads, sector-by-size workbook, census.gov/hfp/btos/data˙downloads, accessed Oct. 3, 2026. [2] U.S. Census Bureau, BTOS AI question wording updates. [3] U.S. Census Bureau, BTOS Methodology, Version 5, Aug. 13, 2026. [4] Hyndman & Athanasopoulos, Forecasting: Principles and Practice, 3rd ed., Σ5.10 (time-series cross-validation).

Code. `python3 analysis.py` reproduces every historical number, table, and figure from the frozen workbook; `prospective.py` reproduces Table 3. **AI assistance.** AI helped write the code and draft and edit the report; the team reviewed the analysis and is responsible for its content.